This document tries to address:
1. Types of interactions and Filtering out residues by interactions using distance ranges.
2. Drawbacks of using the cutoff method.
3. What is Alanine scanning?
4. Theory regarding MD update
5. MD Lysozyme in Water using Complex. pdb
6. Using UCSF chimeraX show better images of complex

1. Intermolecular Forces acting: Hydrogen Bonding, Ion Dipole & Ion induced Dipole, Van Der Walls forces. Van der Walls forces are of 3 types: Keesome, Debye and London Dispersion.

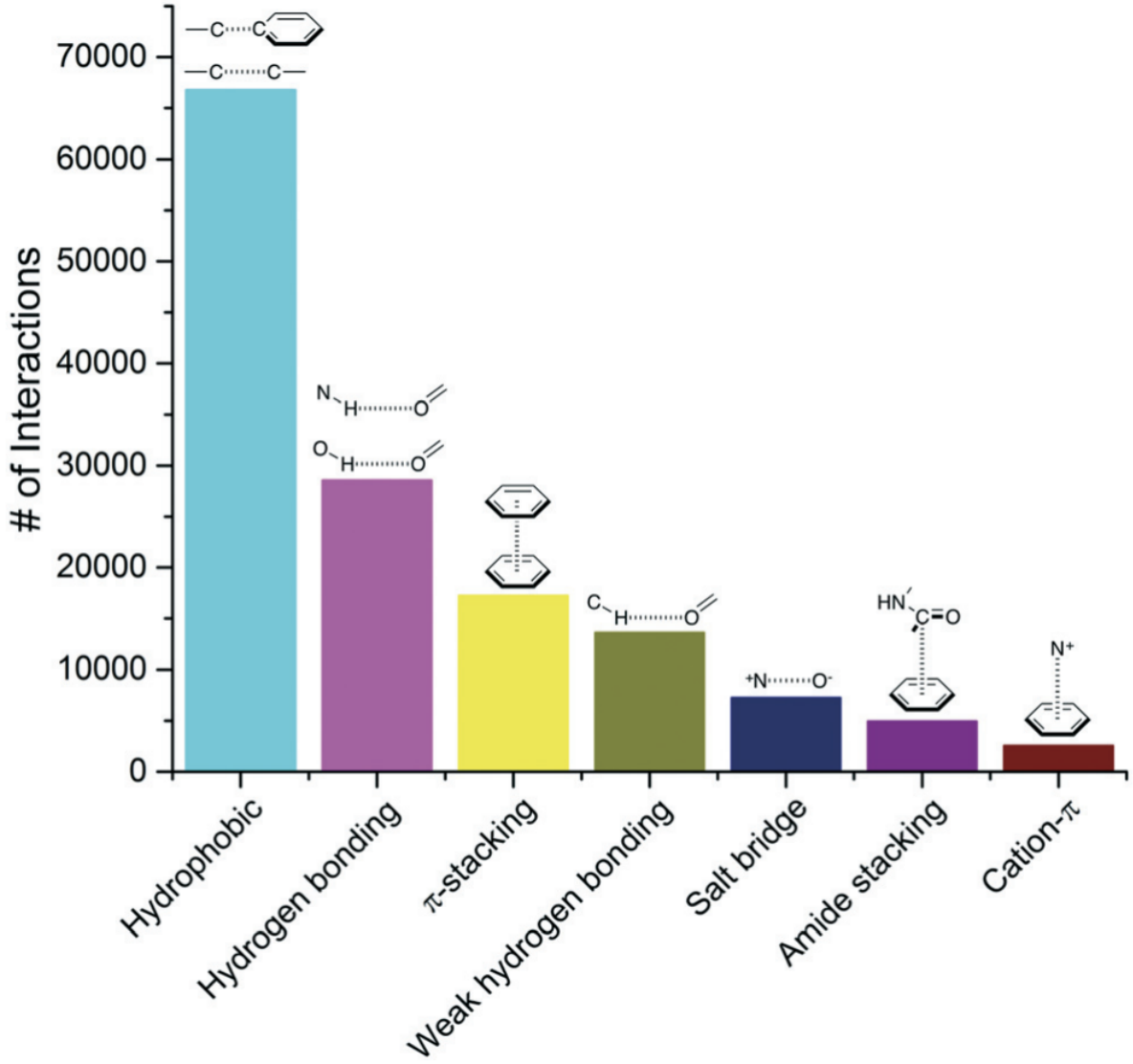


Fig. 1 Frequency distribution of the most common non-covalent interactions observed in protein–ligands extracted from the PDB.

Frequency distribution of the most common non-covalent interactions in protein–ligands extracted from the PDBs REF: (R. F. de Freitasa and M. Schapira, Med. Chem. Commun., 2017, 8, 1970-1981).[(Fig. 1]

2. The drawback of having a cut off 2.75 Armstrong is that we lose a lot of interactions. We also don’t know the type of interaction and we don’t know how significant every residue interaction is.

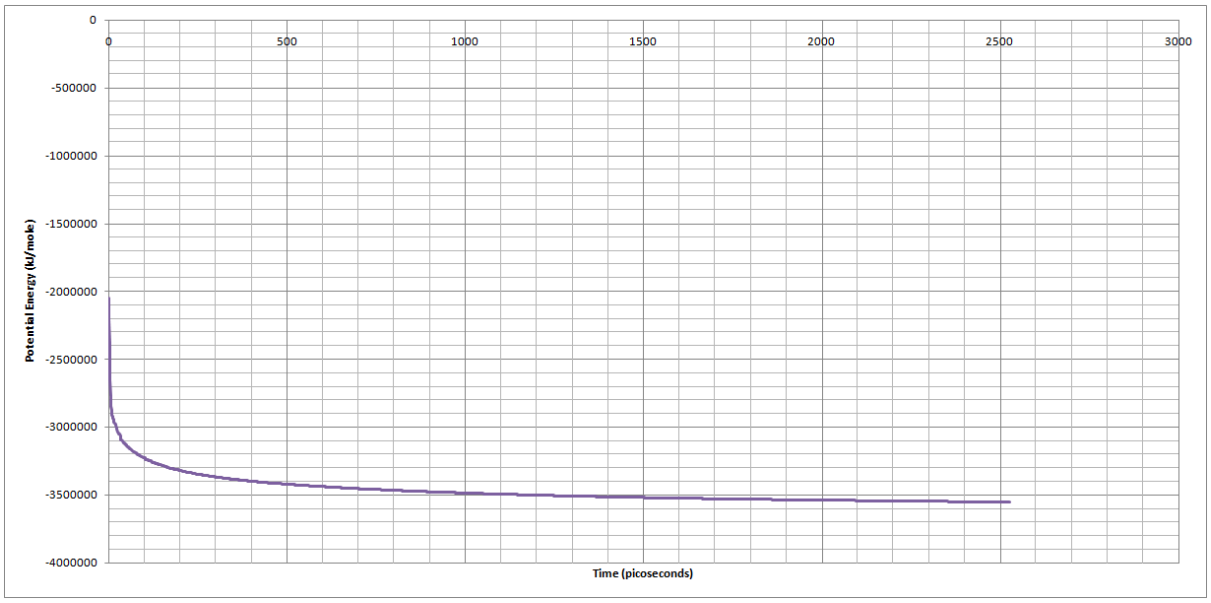
3. Alanine Scanning Mutagenesis: Used to symmetrically substitute specific amino acids with Alanine by altering the codon sequence using standard site-directed in vitro mutagenesis procedures. Alanine is selected because it effectively removes the side chain functionality without introducing any flexibility. Glycine is not used as it has 2 Hydrogens and it adds on flexibility which will change the structure. Hence the next larger amino acid Alanine is selected. Residues which show weaker binding are more important and hence significant.

Binomial Alanine scanning: Amino acids can interact with one another in both additive and synergistic ways. For example, 2 amino acids can form a salt bridge but if only one is mutated into an amino acid, we might not be able to observe its effect. The idea here is to make multiple mutations into alanine at once and look at the protein function.

4. For theory regarding MD, I am referring to the book by Daan Frenkel and Berend Smith. There is also a good paper by Scott A. Hollingsworth and Ron Dror from Stanford University titled Molecular Dynamics Simulations for All. I read about the contribution of Rahman(Aneesur Rahman) to Molecular Dynamics simulations through his work on the water molecule and his work on Argon using Leonard Jones' Potential. While my knowledge is rudimentary and needs work, I think I made some good first steps.

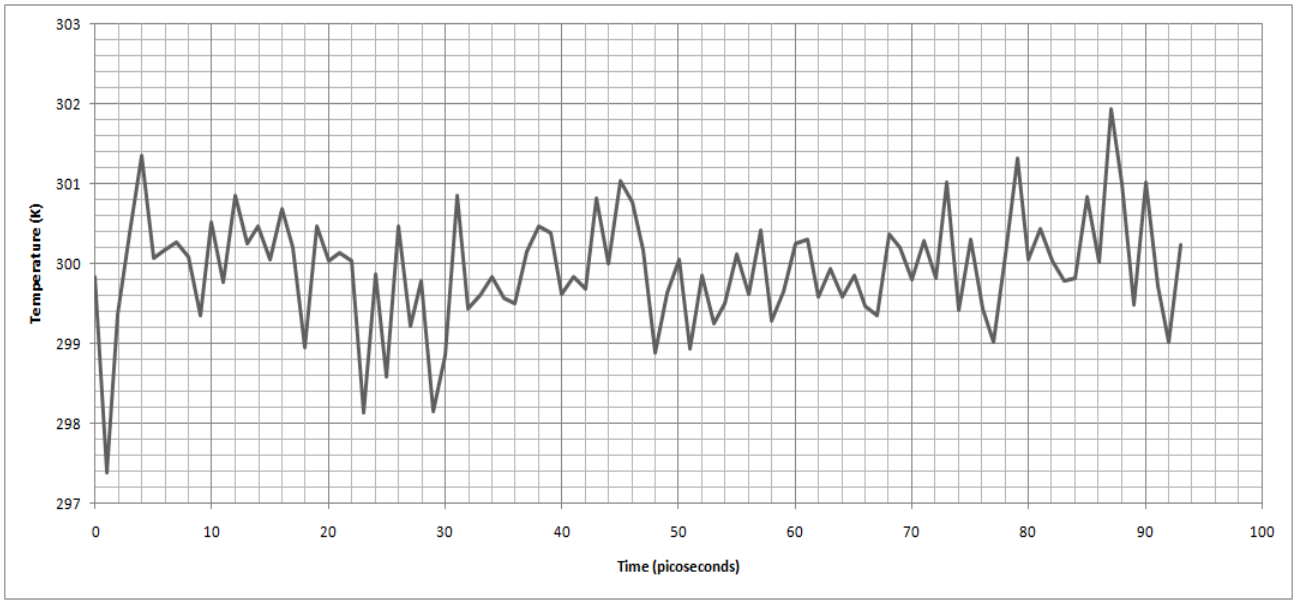
5. Results of Molecular Dynamics Simulations :

The solvated, electroneutral system is assembled. Before we can begin dynamics, we must ensure that the system has no steric clashes or inappropriate geometry. The structure is relaxed through a process called energy minimization (EM).



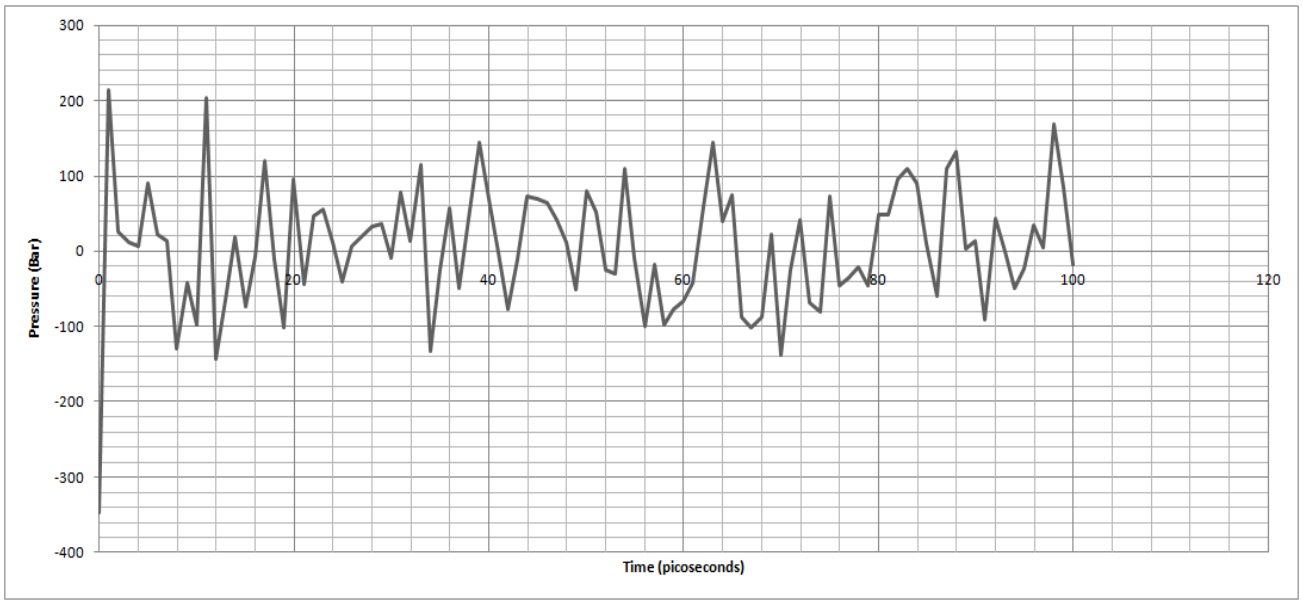
Potential Energy of the Complex (kJ/mole) with Time(ps)

EM ensured that we have a reasonable starting structure, in terms of geometry and solvent orientation.



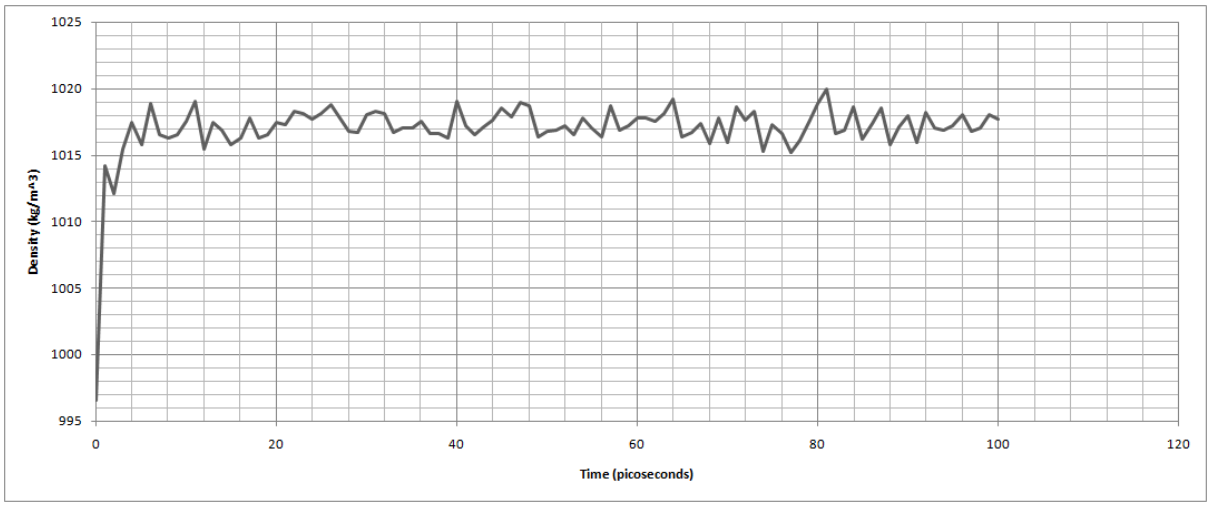
Temperature (K) vs Time (ps) for Complex

From the plot, it is clear that the temperature of the system quickly reaches the target value (300 K), and remains stable over the remainder of the equilibration.



Pressure(bar) vs Time(ps) for Complex

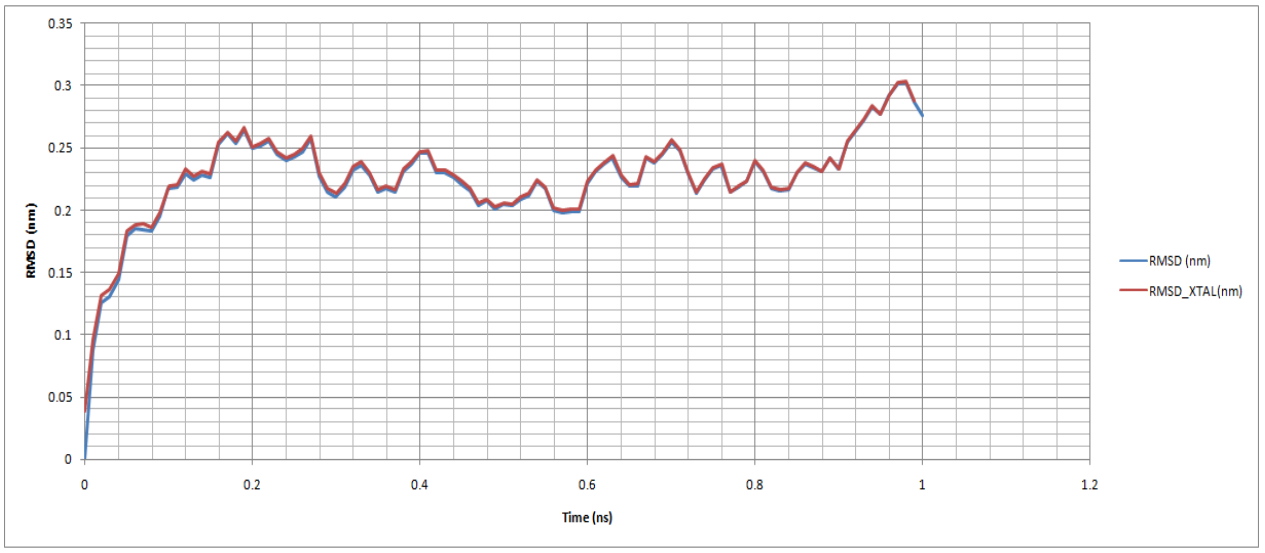
The pressure value fluctuates widely over the course of the 100-ps equilibration phase, but this behavior is not unexpected. Reference pressure was set to 1 bar, so is this outcome acceptable. Pressure is a quantity that fluctuates widely over the course of an MD simulation.



Density(kg/m^3) vs Time(ps) for Complex

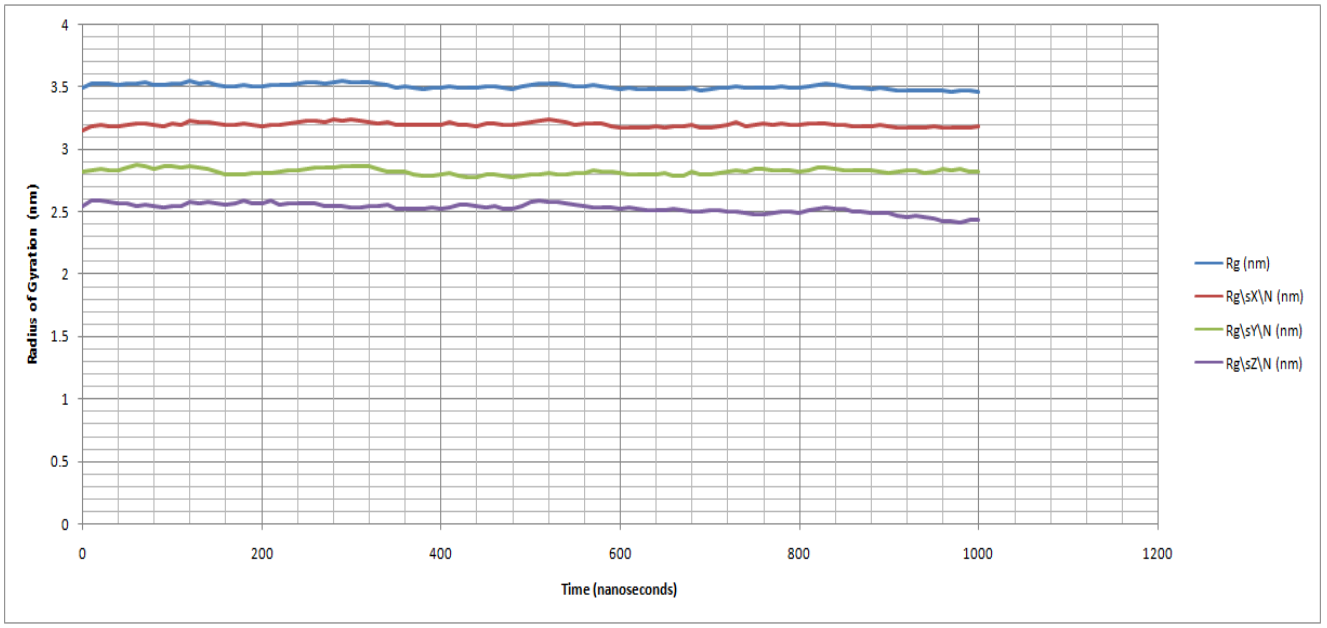
The density values are very stable over time, indicating that the system is well-equilibrated now with respect to pressure and density.

Production MD faced issues as I have limited hardware and was unable to recompile GMX for GPU capabilities. It took close to 1 day 2 hours and 30 mins to run the Production MD.



RMSD (nm) vs Time (ns) for both Equilibrated and Crystal

Both time series show the RMSD levels off to ~0.1 nm (1 Å), indicating that the structure is very stable. Subtle differences between the plots indicate that the structure at t = 0 ns is slightly different from this crystal structure. This is to be expected, since it has been energy-minimized, and because the position restraints are not 100% perfect, as discussed previously.



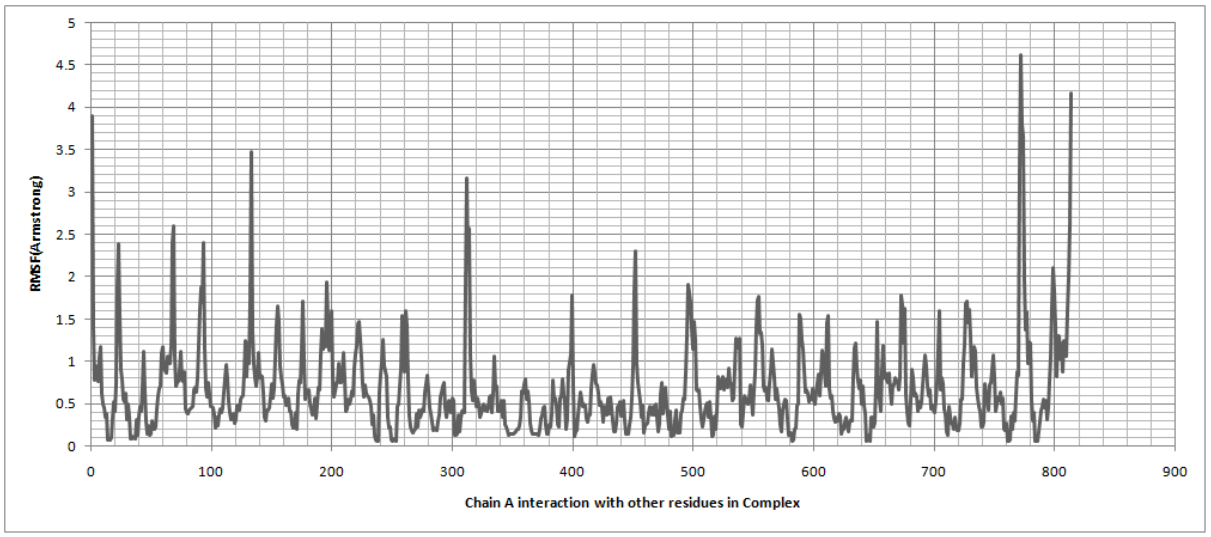
Radius of Gyration (nm) vs Time (ns)

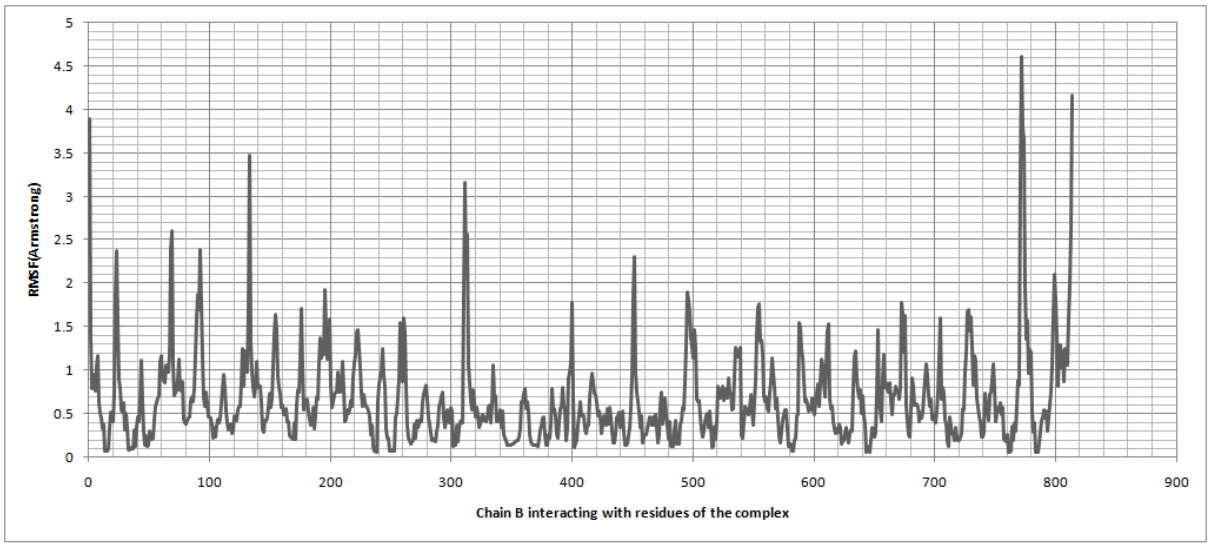
The radius of gyration of a protein is a measure of its compactness. If a protein is stably folded, it will likely maintain a relatively steady value of R_g. If a protein unfolds, its R_g will change over time.

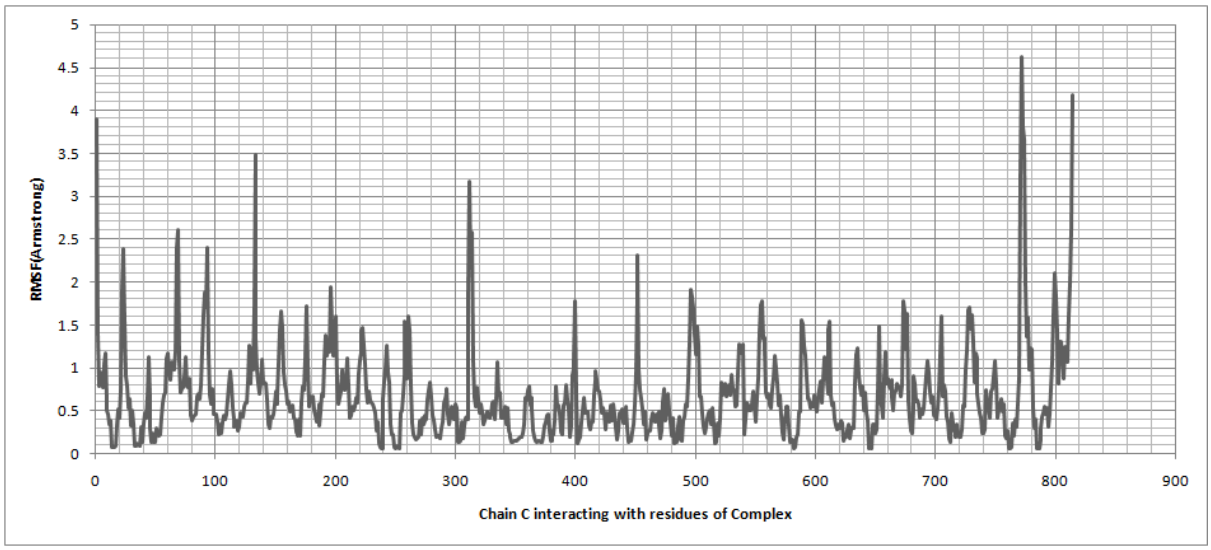
We can see from the reasonably invariant R_g values that the protein remains very stable, in its compact (folded) form over the course of 1 ns at 300 K. This result is not unexpected, but illustrates an advanced capacity of GROMACS analysis that comes built-in.

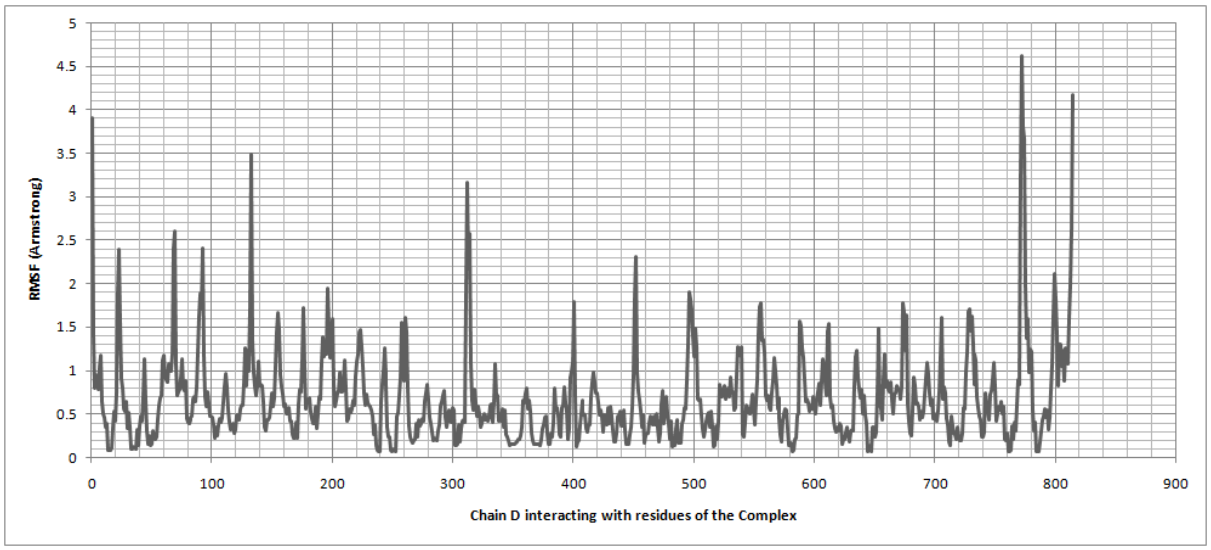
Using a Server I have tried to Generate the RMSF values for every chain. I used the CABS-flex2.0 server for my job.

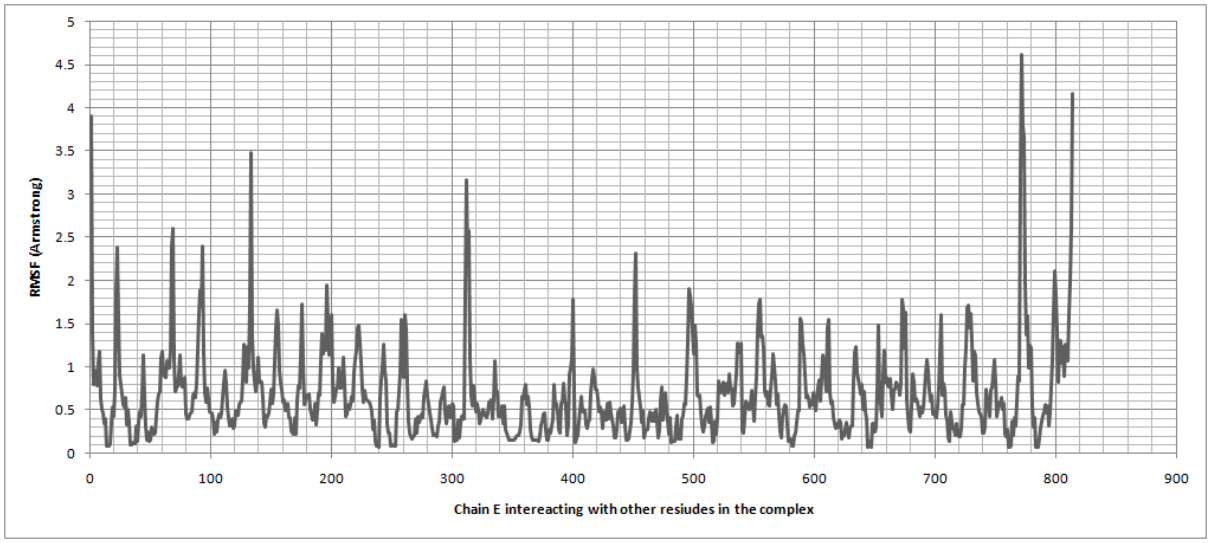
Protein rigidity	1.0
Protein restraints	ss2 3 3.8 8.0
Global C-alpha restraints weight	1.0
Global Side-chain restraints weight	1.0
Number of cycles	50
Cycles between trajectory	50
Temperature range	1.40 1.40
RNG seed	3380

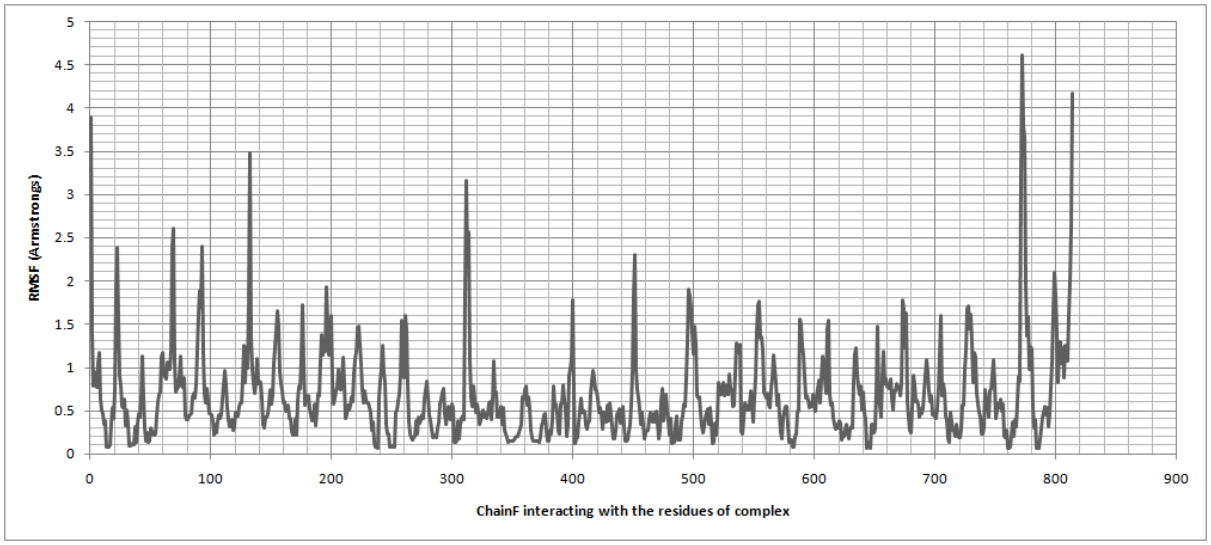

RMSF(Armstrong)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500
600
700
800
900
Chain A interaction with other residues in Complex

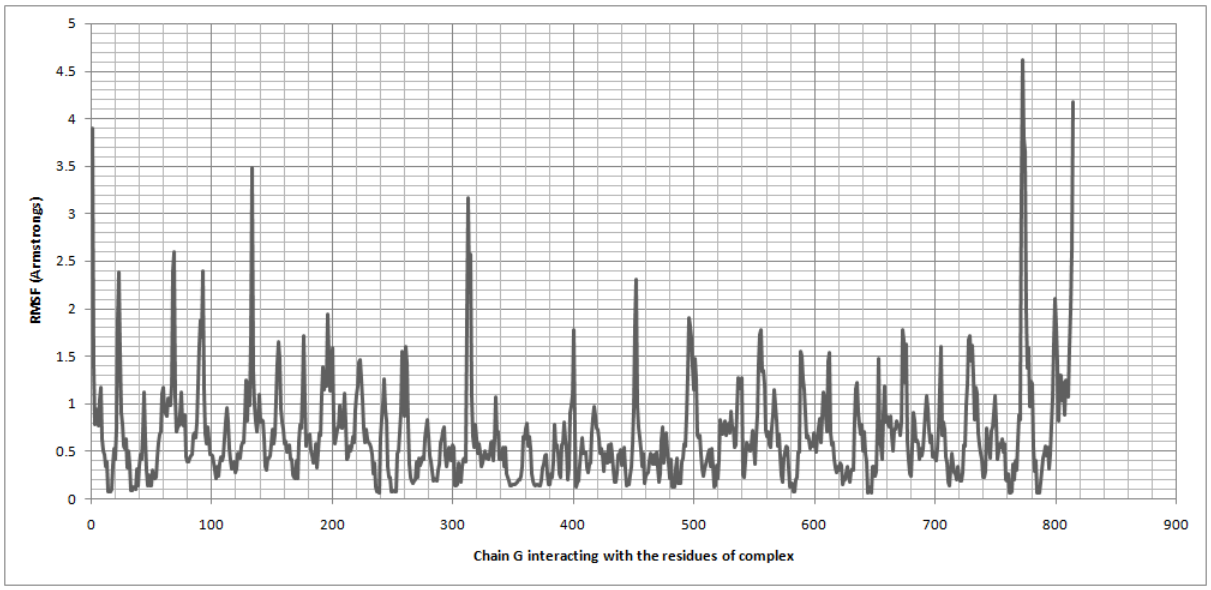

RMSF(Armstrong)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500
600
700
800
900
Chain B interacting with residues of the complex

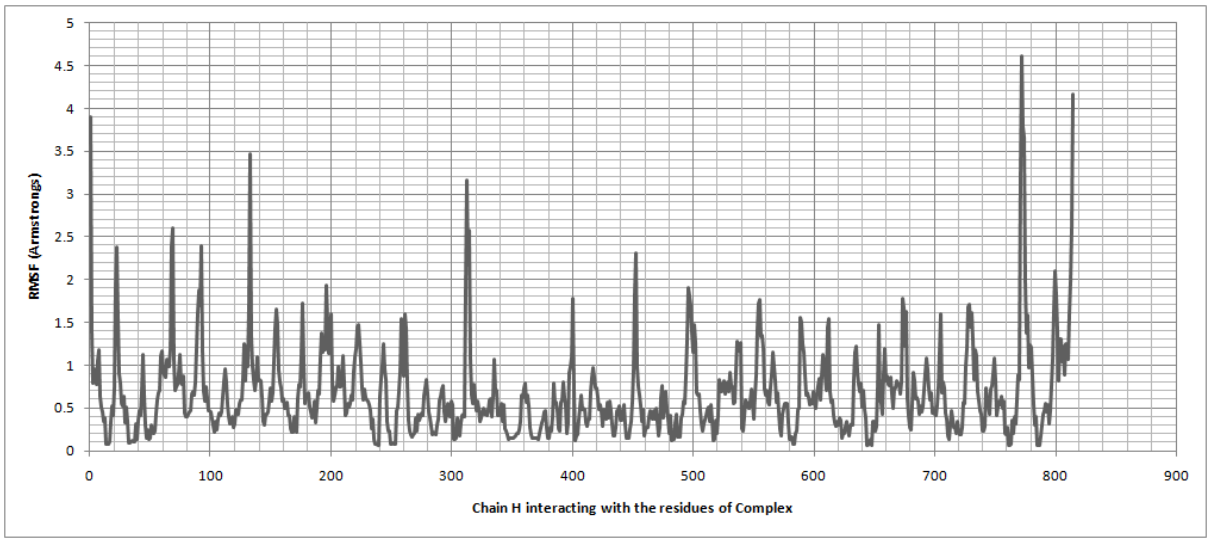

RMSF(Armstrong)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500
600
700
800
900
Chain C interacting with residues of Complex

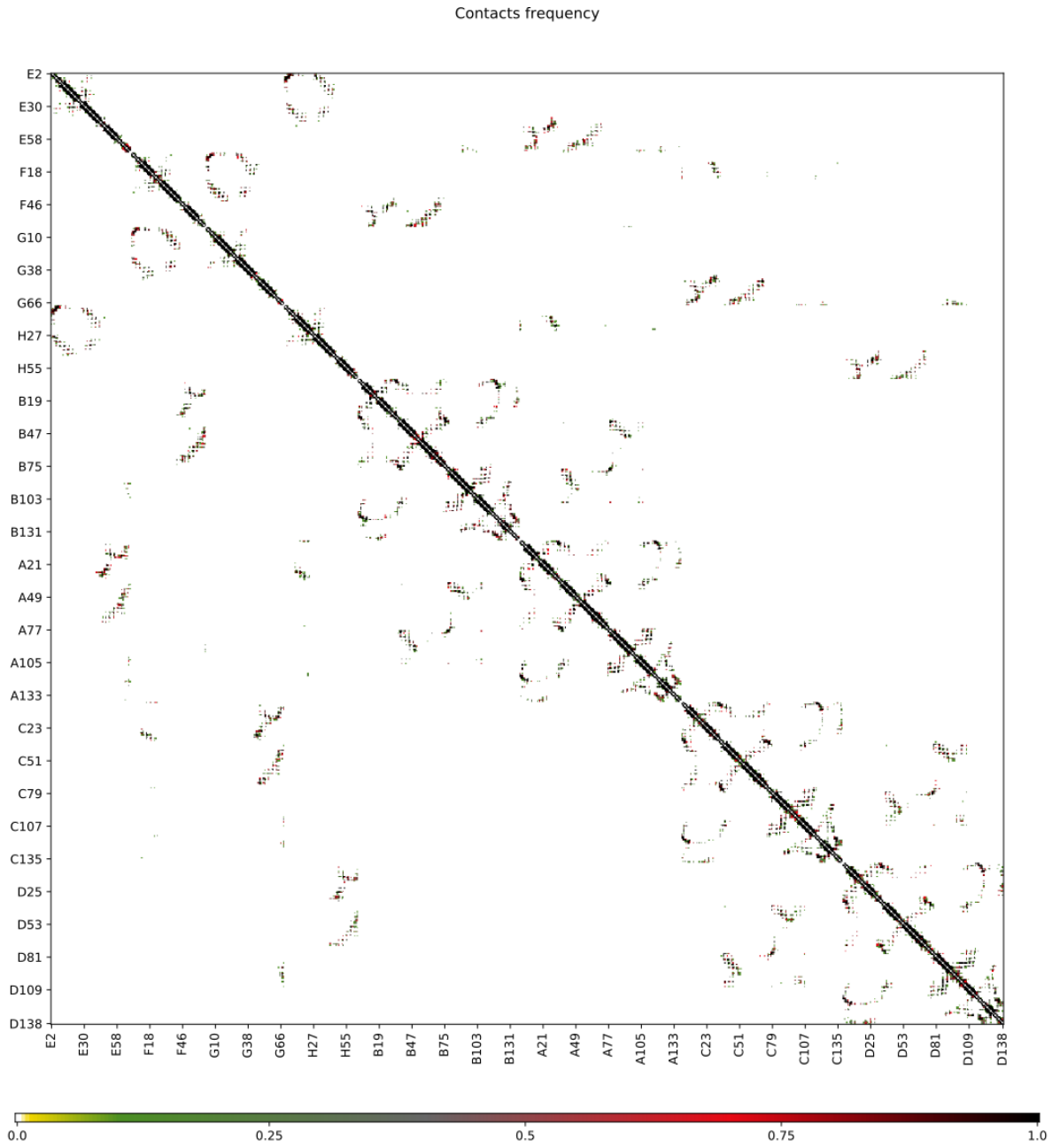

RMSF (Armstrong)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500
600
700
800
900
Chain D interacting with residues of the Complex


RMSF (Armstrong)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500
600
700
800
900
Chain E intereacting with other resiudes in the complex


RMSF (Armstrongs)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500
600
700
800
900
ChainF interacting with the residues of complex


RMSF (Armstrongs)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500
600
700
800
900
Chain G interacting with the residues of complex

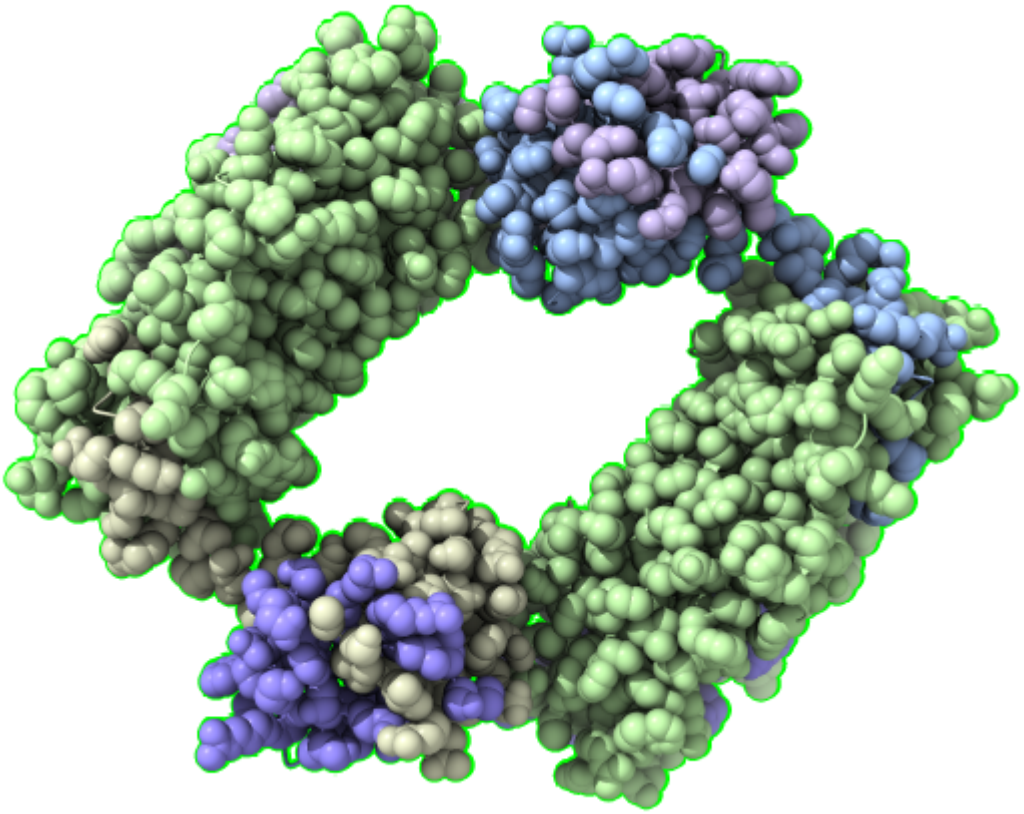

RMSF (Armstrongs)
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500
600
700
800
900
Chain H interacting with the residues of Complex

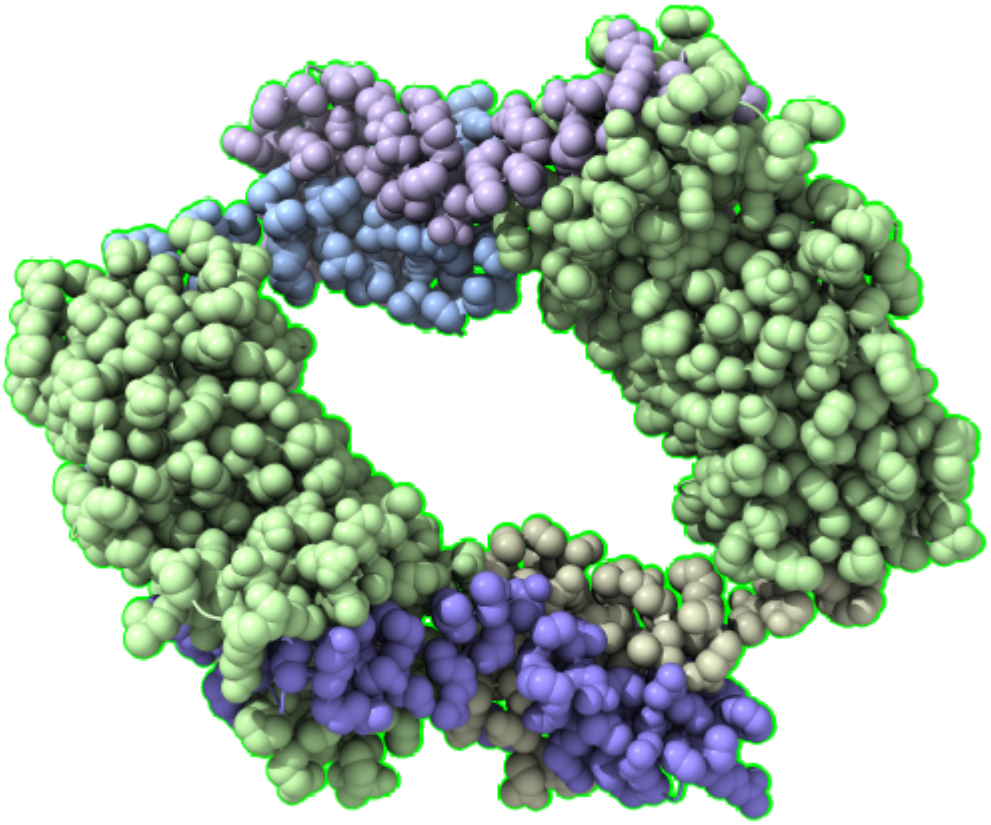


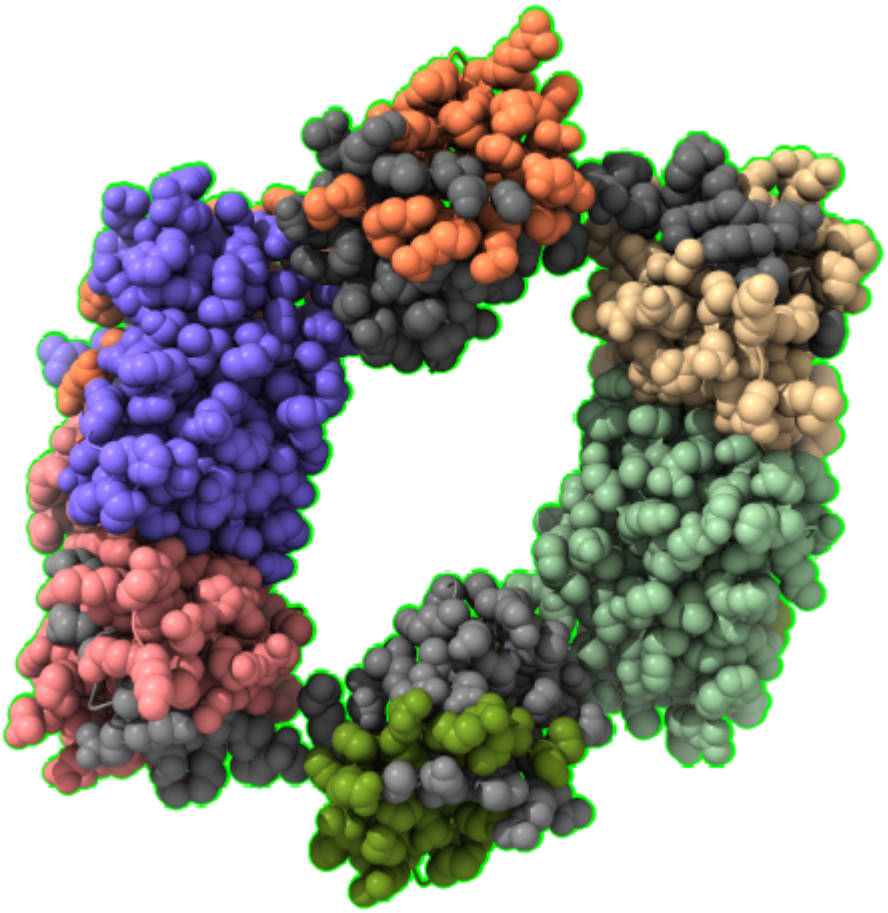
Contact Map

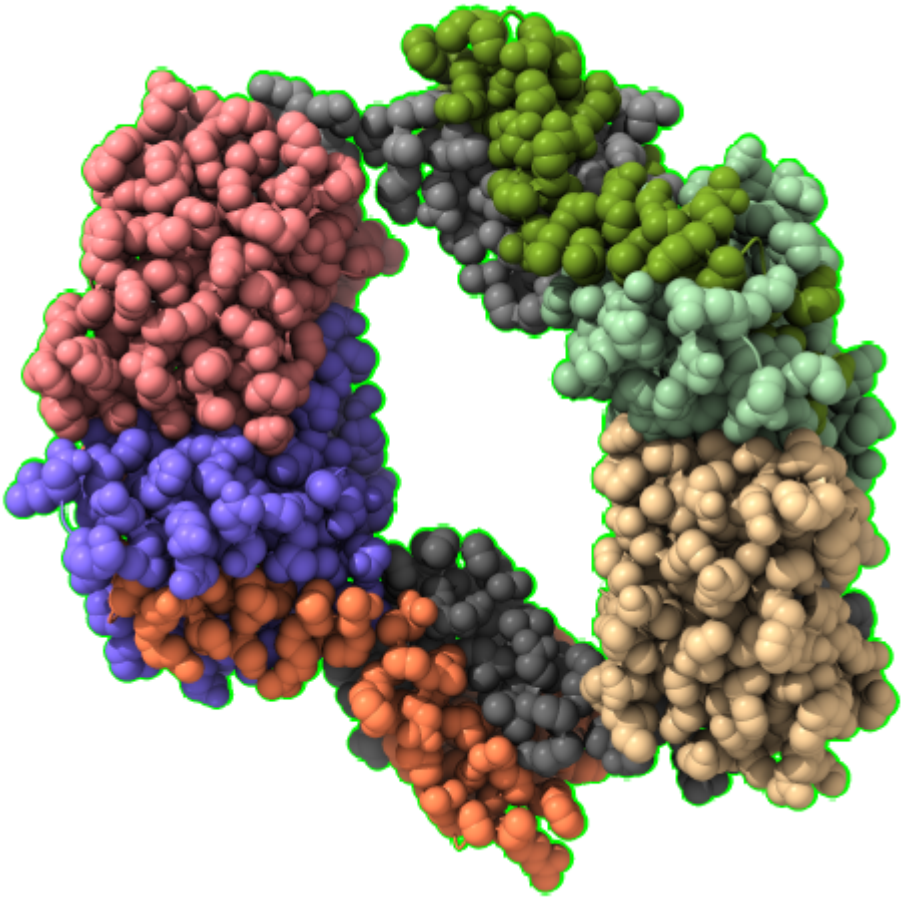
A protein contact map represents the distance between all possible amino acid residue pairs of a three-dimensional protein structure using a binary two-dimensional matrix. For two residues, the element of the matrix is 1 if the two residues are closer than a predetermined threshold, and 0 otherwise. Various contact definitions have been proposed: The distance between the C_α-C_α atom with a threshold of 6-12 Å; the distance between C_β-C_β atoms with a threshold of 6-12 Å (C_α is used for Glycine); and the distance between the side-chain centers of mass.

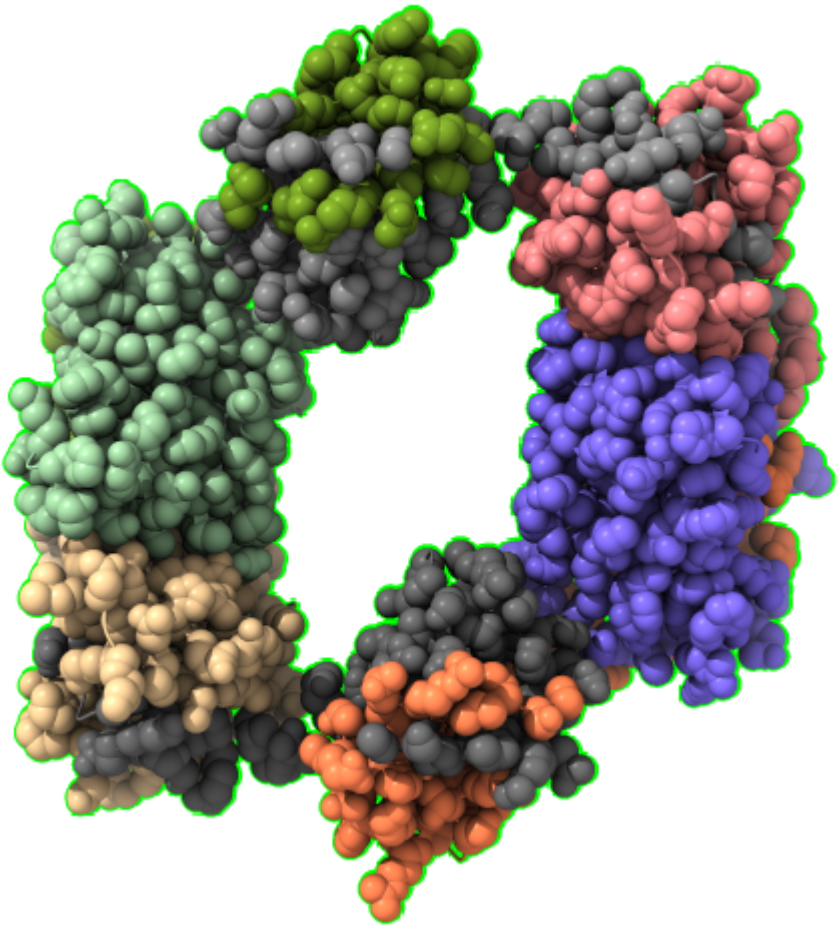
6. I used UCSF ChimeraX to make images. I saw a lot of papers which used this software and this made a lot of images. I used various styles and have seen some being from many popular papers. I played with lighting, colours, styles and selections.

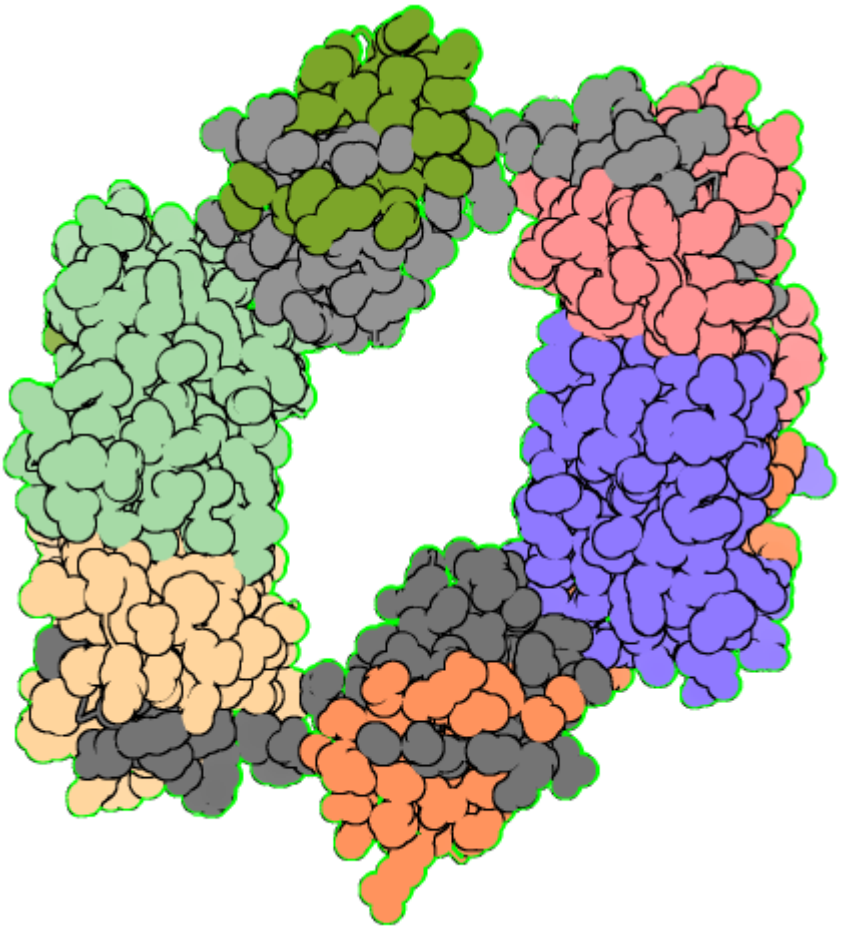


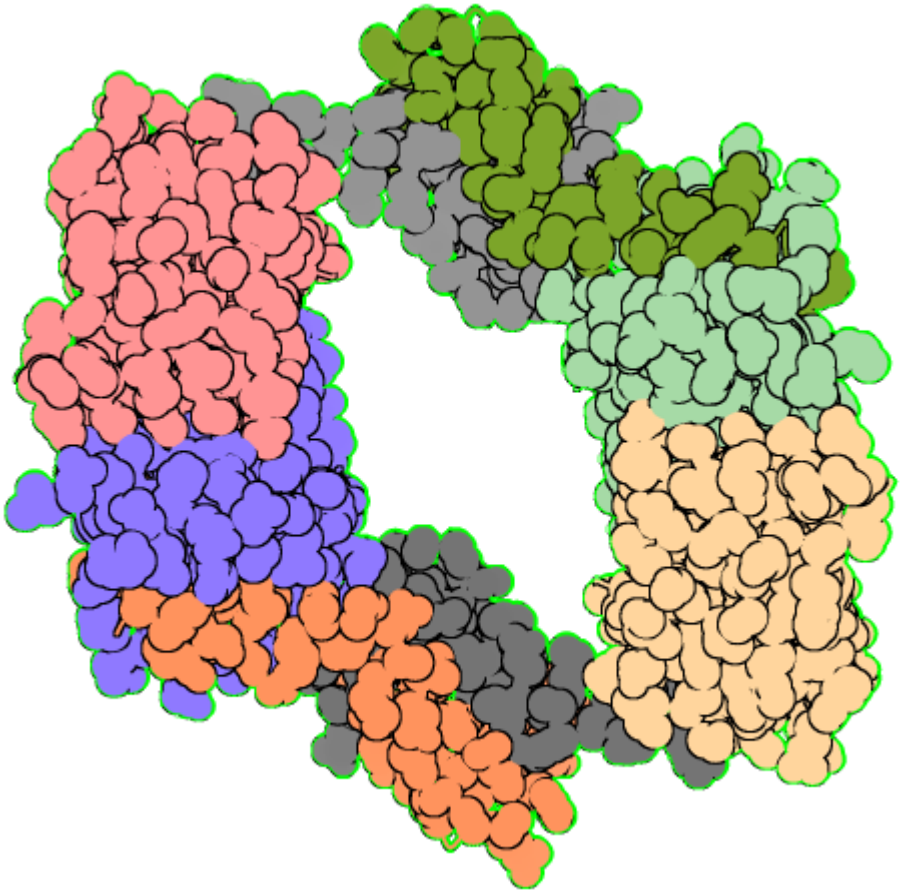


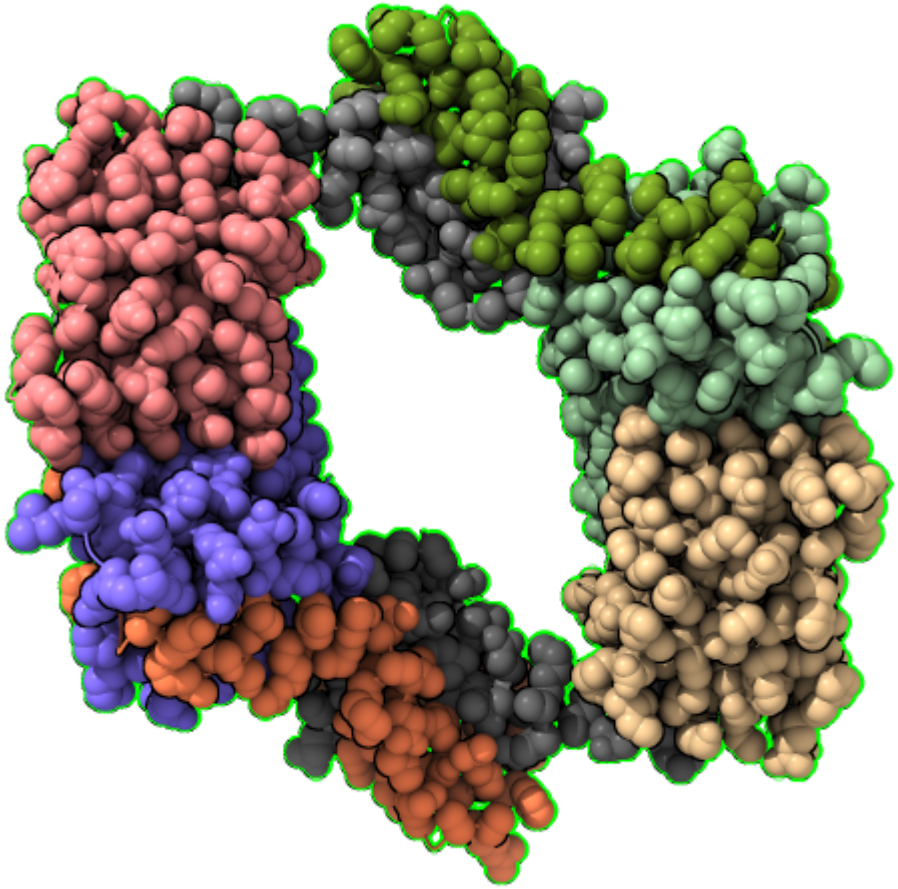


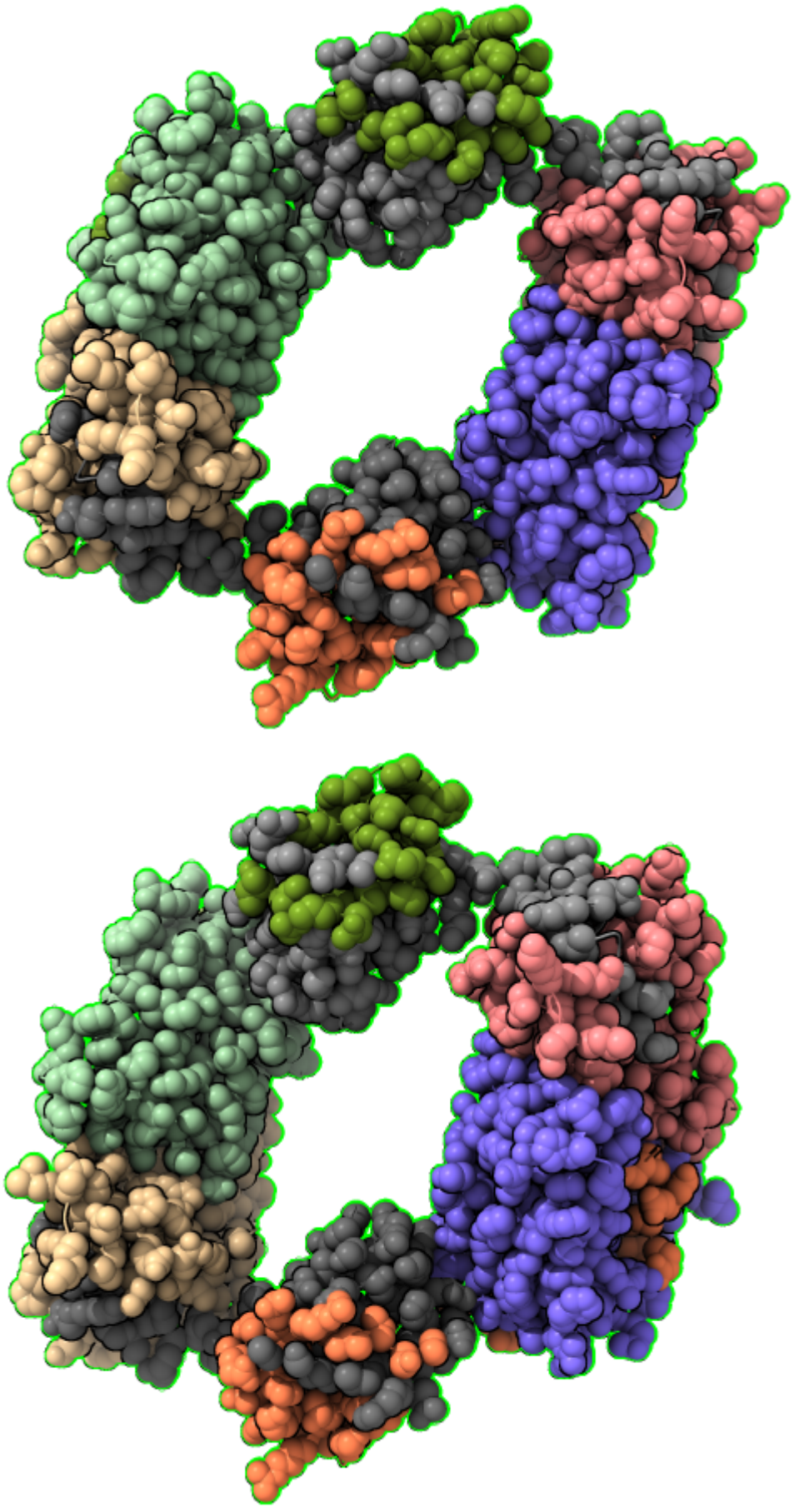


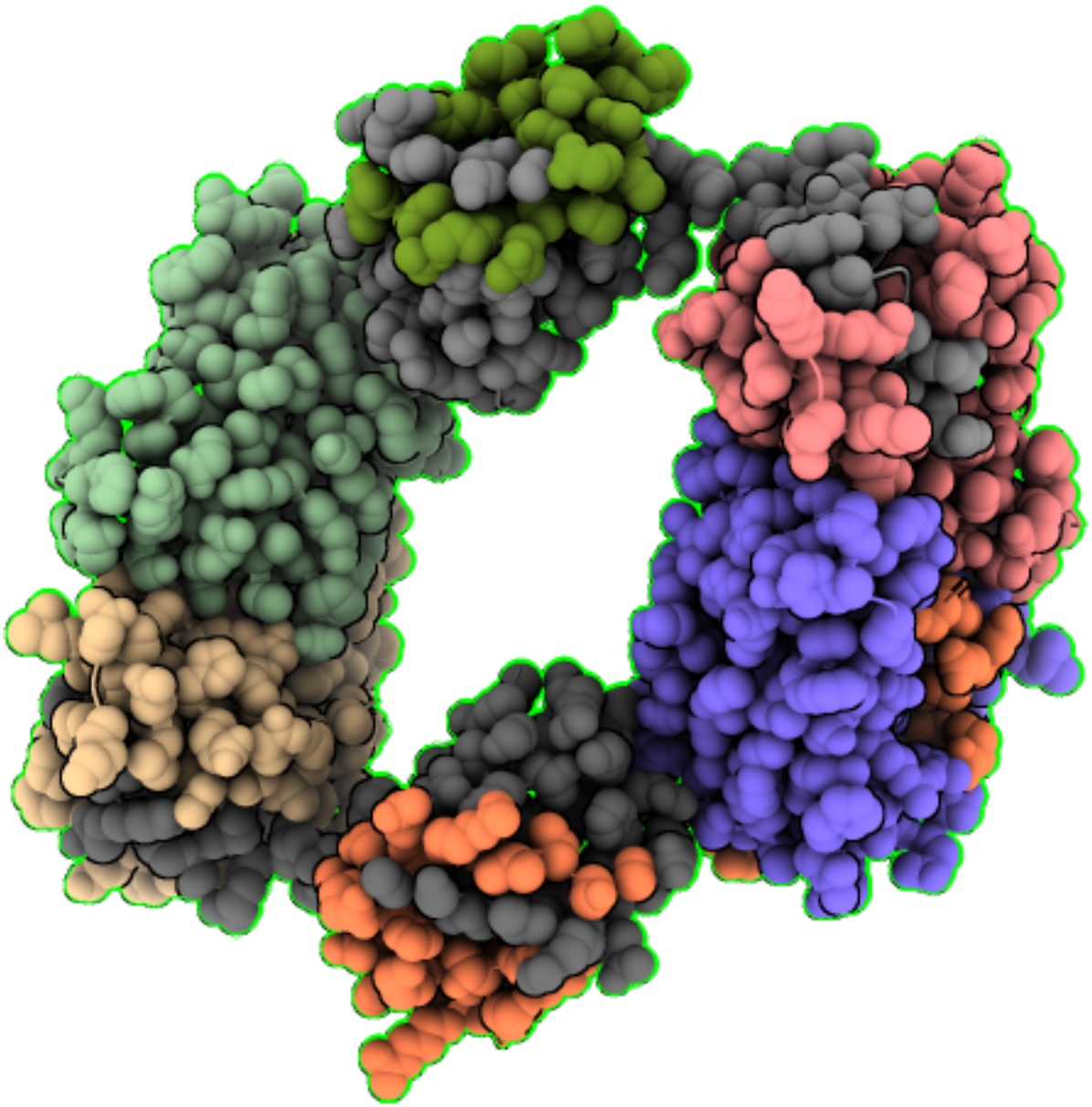


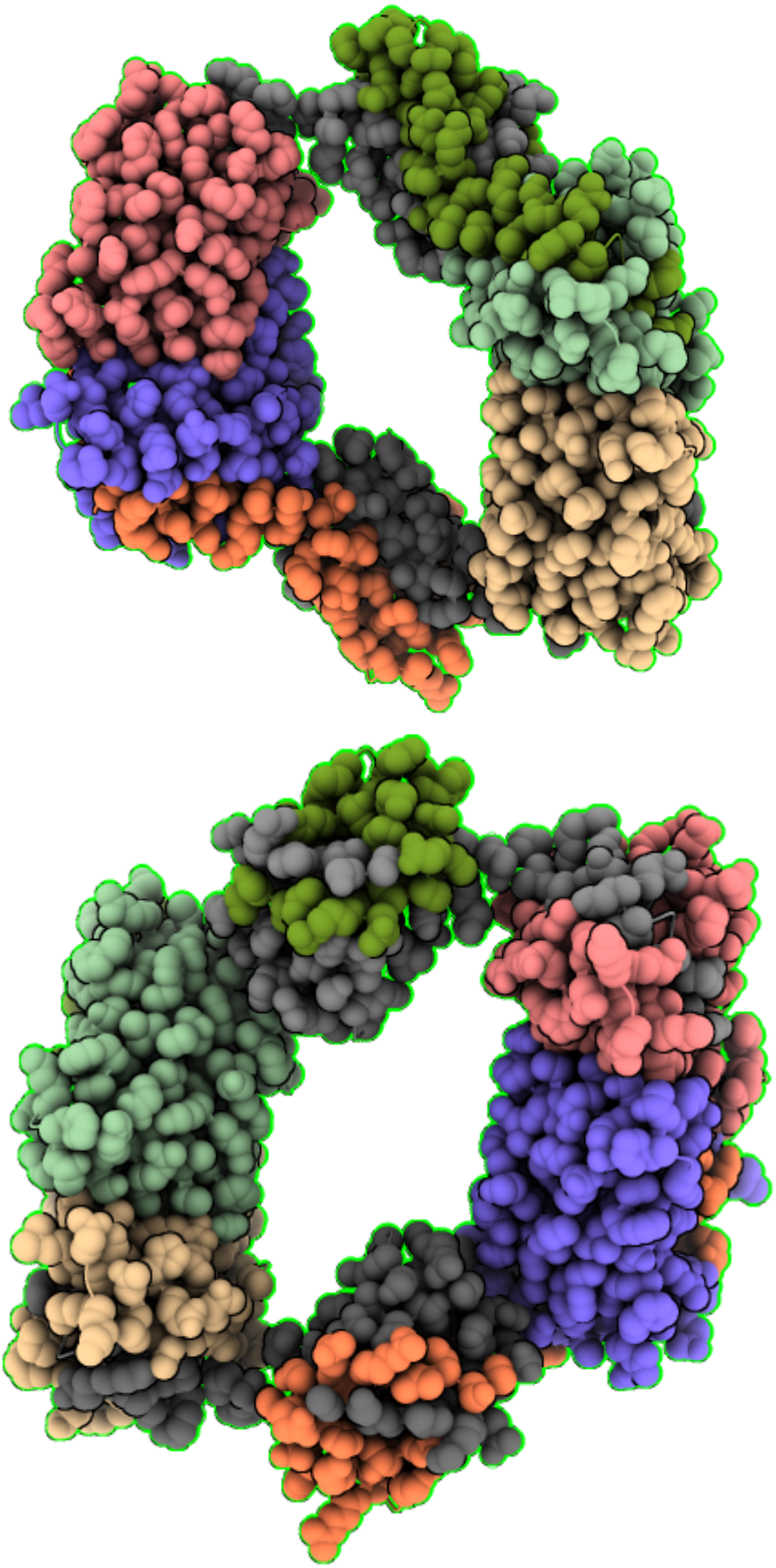












References:

3. https://pubs.acs.org/doi/pdf/10.1021/ar020082r
4. Hollingsworth SA, Dror RO. Molecular Dynamics Simulation for All. Neuron. 2018 Sep 19;99(6):1129-1143. doi: 10.1016/j.neuron.2018.08.011. PMID: 30236283; PMCID: PMC6209097.
5. J.A. Lemkul (2018) "From Proteins to Perturbed Hamiltonians: A Suite of Tutorials for the GROMACS-2018 Molecular Simulation Package, v1.0" *Living J. Comp. Mol. Sci.* 1 (1): 5068.
6. UCSF ChimeraX: Structure visualization for researchers, educators, and developers. Pettersen EF, Goddard TD, Huang CC, Meng EC, Couch GS, Croll TI, Morris JH, Ferrin TE. *Protein Sci.* 2021 Jan;30(1):70-82.
7. UCSF ChimeraX: Meeting modern challenges in visualization and analysis. Goddard TD, Huang CC, Meng EC, Pettersen EF, Couch GS, Morris JH, Ferrin TE. *Protein Sci.* 2018 Jan;27(1):14-25.
8.Kuriata, A.*, Gierut, A. M.*, Oleniecki, T., Ciemny, M. P., Kurcinski, M., Kmiecik, S. (2018). CABS-flex 2.0: a web server for fast simulations of the flexibility of protein structures. Nucleic acids research.
9. R. F. de Freitas and M. Schapira, Med. Chem. Commun., 2017, 8, 1970-1981